Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 10
timesteps_orig: 60
timesteps: 6
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.1
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
forecast_steps: 20

LSTM Multi-Step Training Report

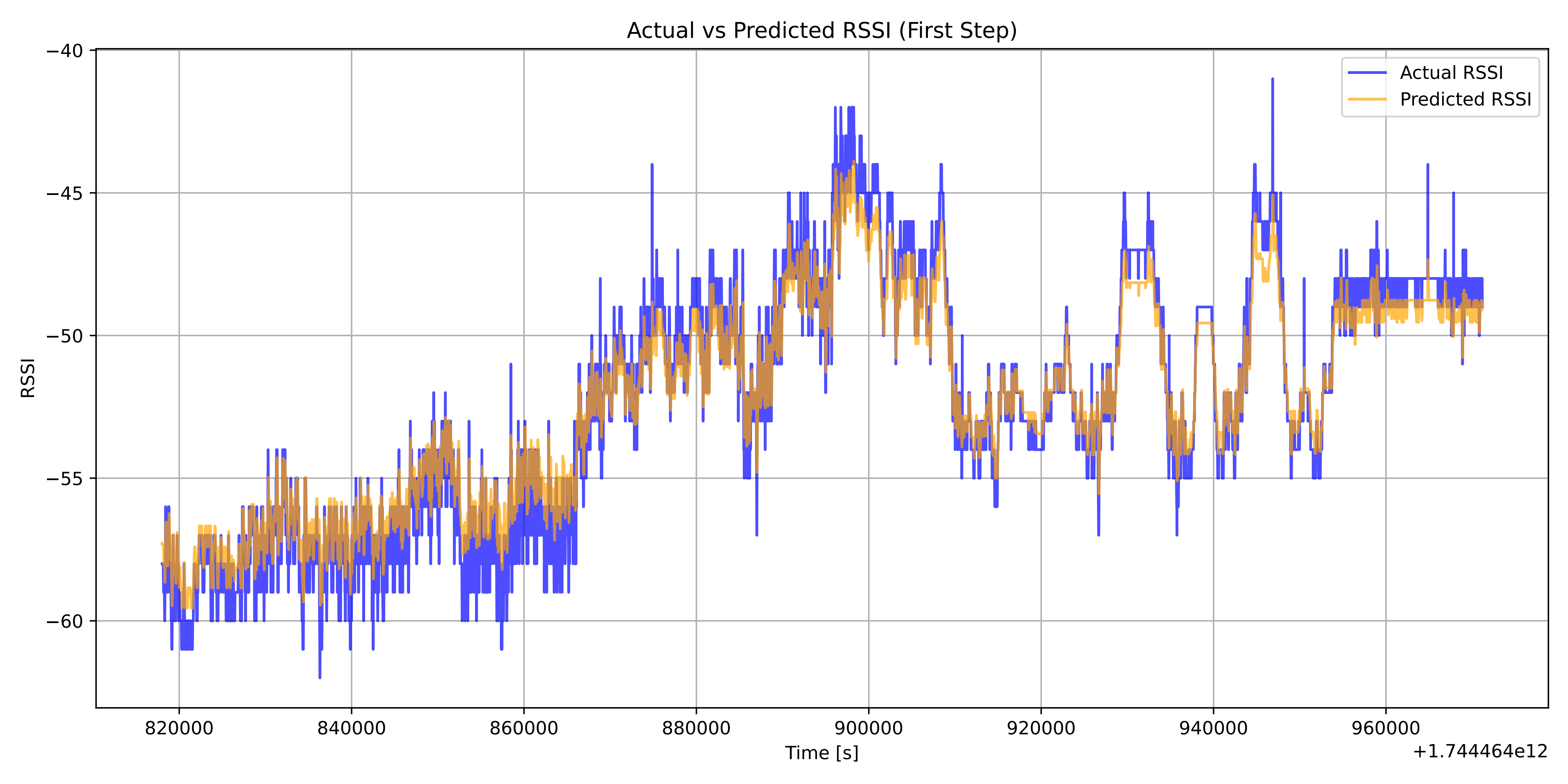
Report Generated: 2025-05-10 12:09:34

Training Start: 2025-05-10_11-08-29
Training End: 2025-05-10_11-18-55
Training Duration: 10.43 minutes

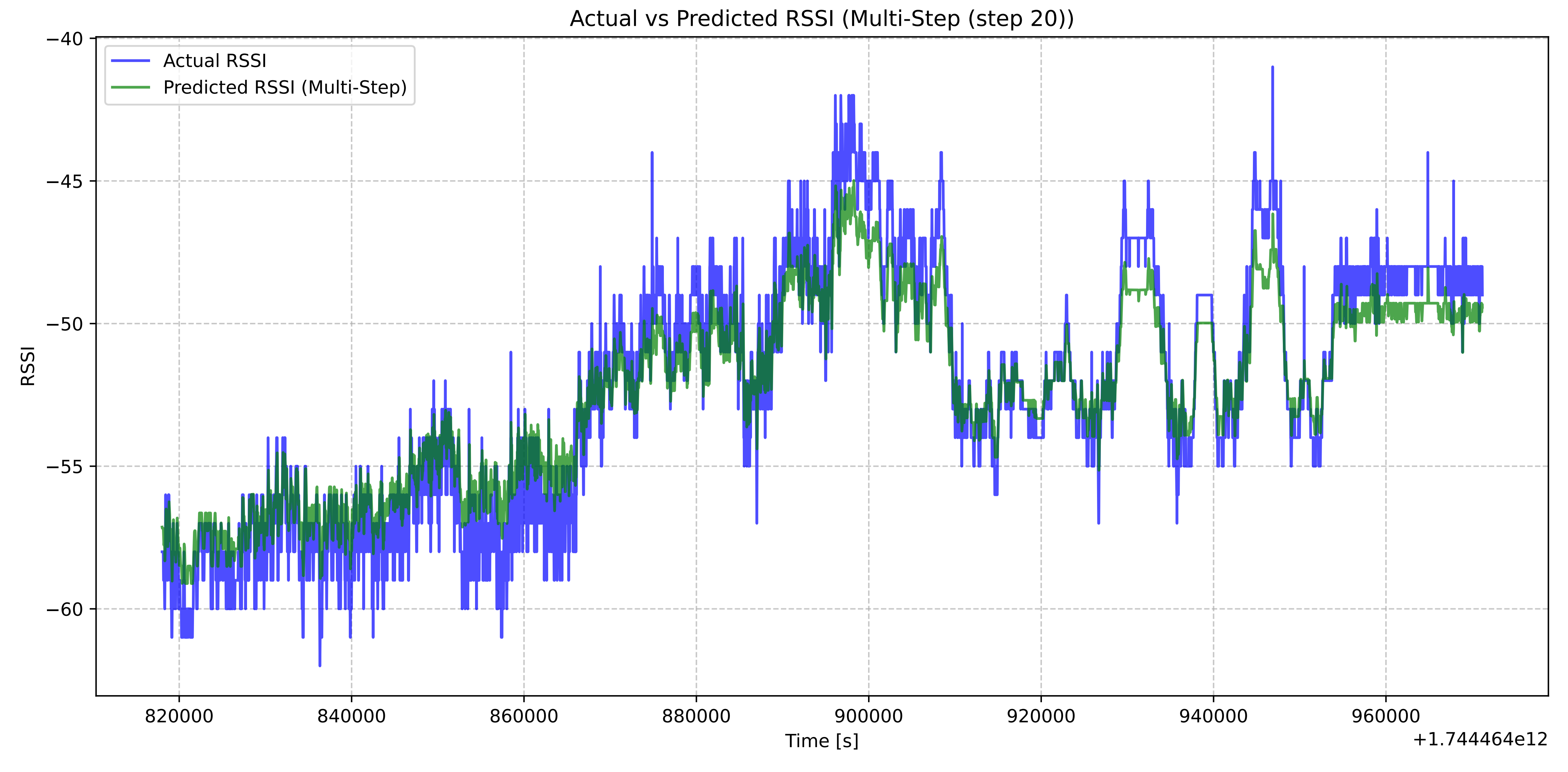
Best Hyperparameters:
Number of Layers: 2
Activation: tanh
Optimizer: rmsprop
Dropout Rate: 0.20

MAE: 1.1452
MSE: 2.0445
RMSE: 1.4299
R2 Score: 0.8875

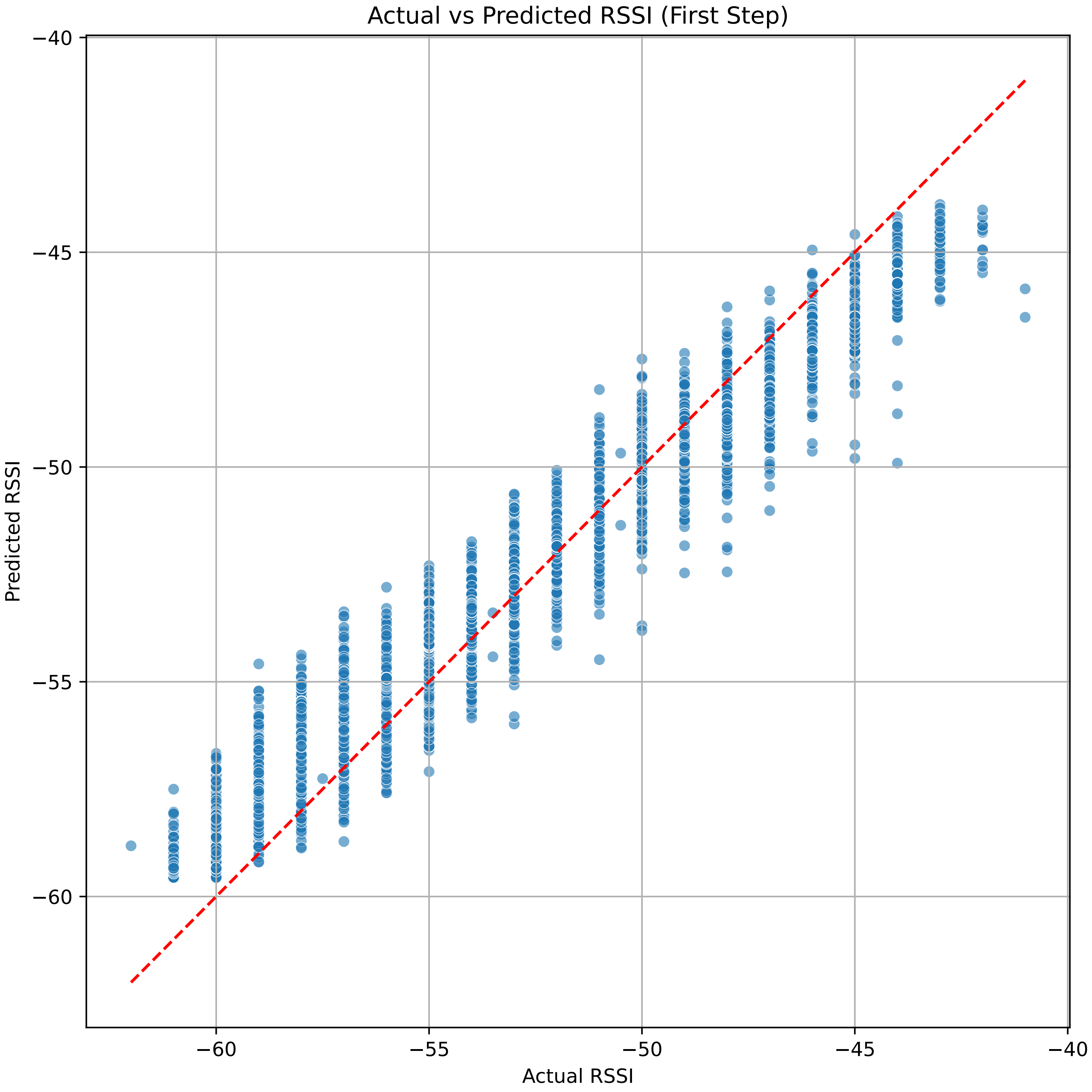
Multi-Step Forecast:
Forecast Steps: 20
MAE: 0.0124
MSE: 0.0002
RMSE: 0.0155
R2 Score: 0.9407

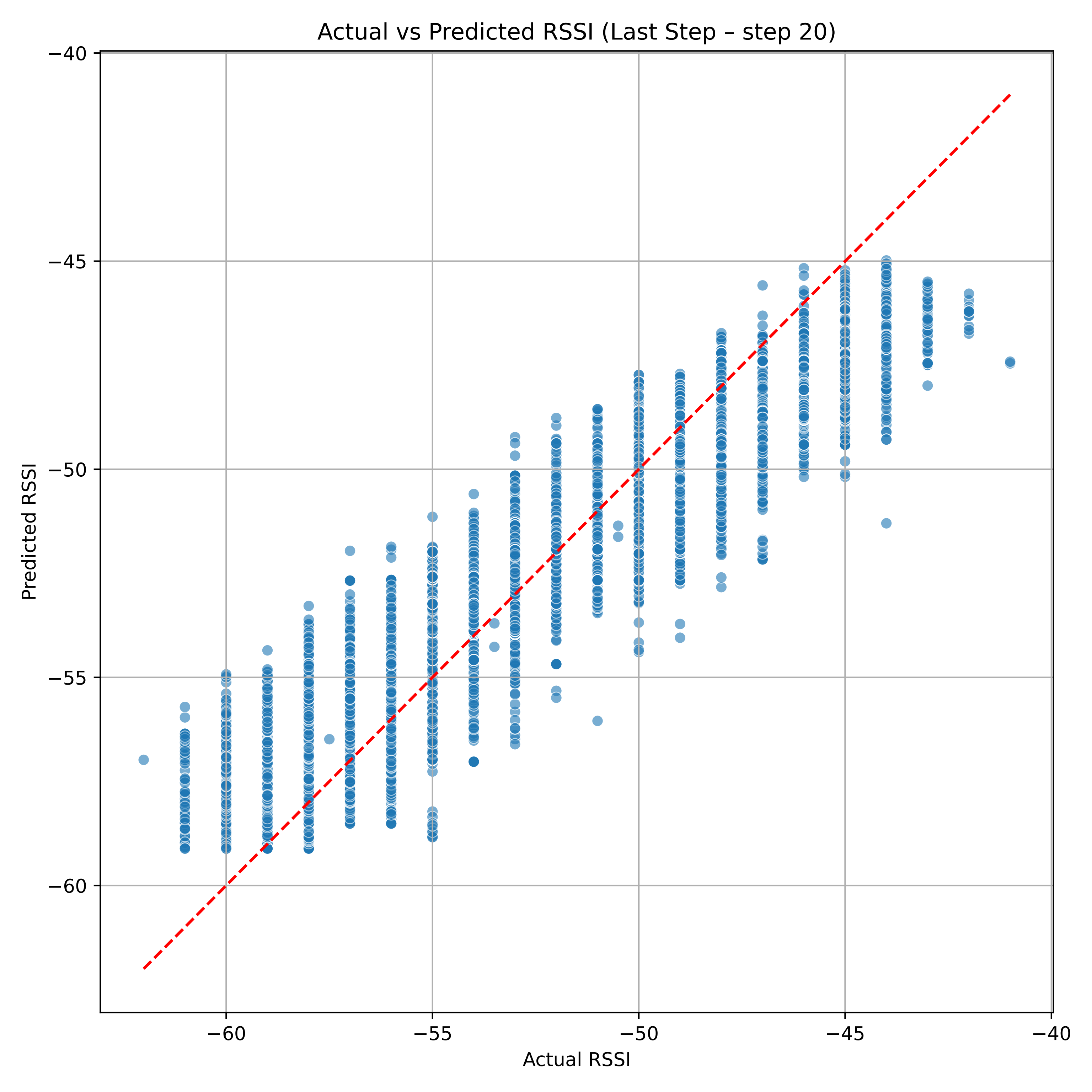


Actual vs Predicted RSSI (First Step)
Actual RSSI
Predicted RSSI
RSSI
−40
−45
−50
−55
−60
820000
840000
860000
880000
900000
920000
940000
960000
Time [s]
+1.744464e12

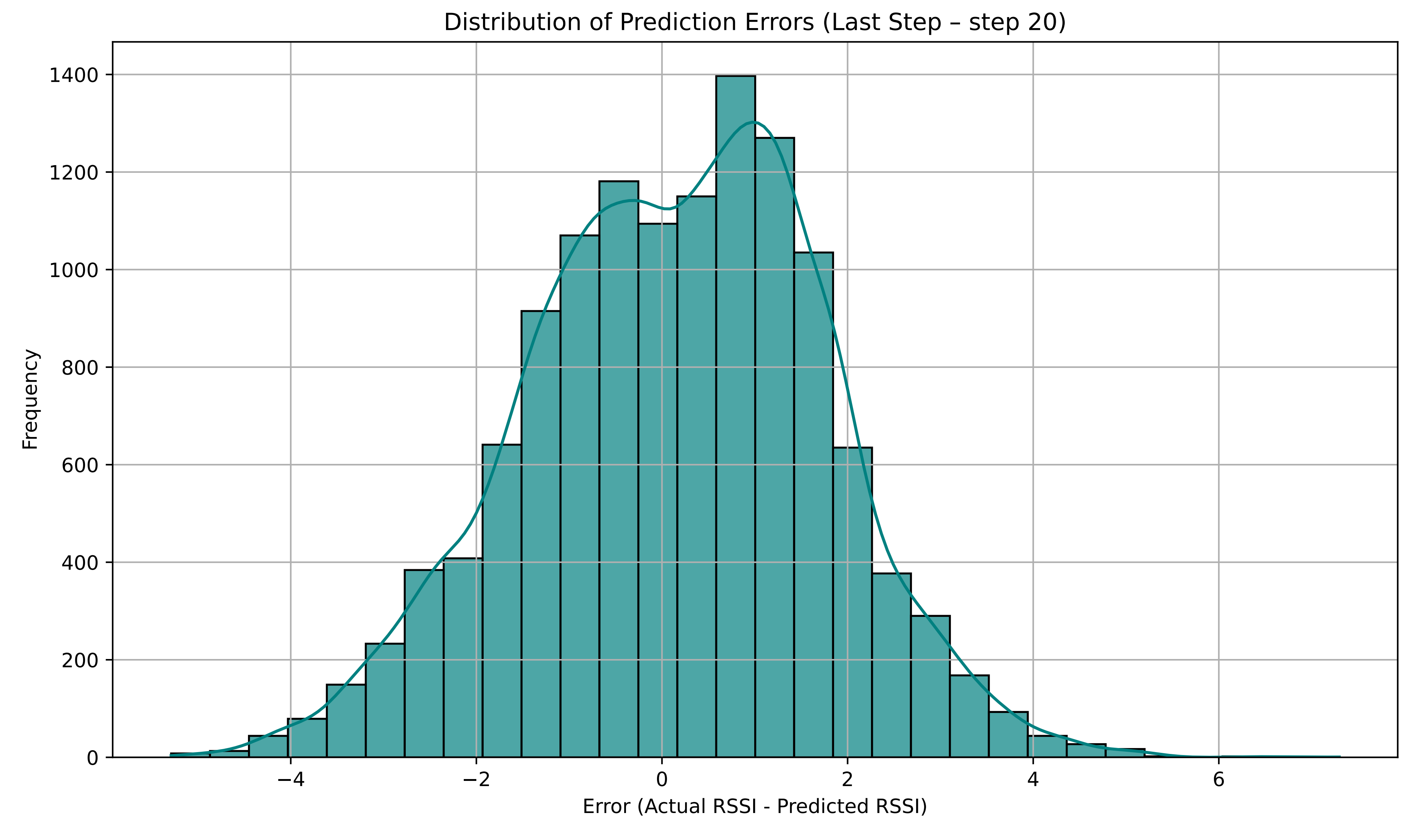


Actual vs Predicted RSSI (Multi-Step (step 20))
Actual RSSI
Predicted RSSI (Multi-Step)
−40
−45
−50
−55
−60
RSSI
820000
840000
860000
880000
900000
920000
940000
960000
Time [s]
+1.744464e12

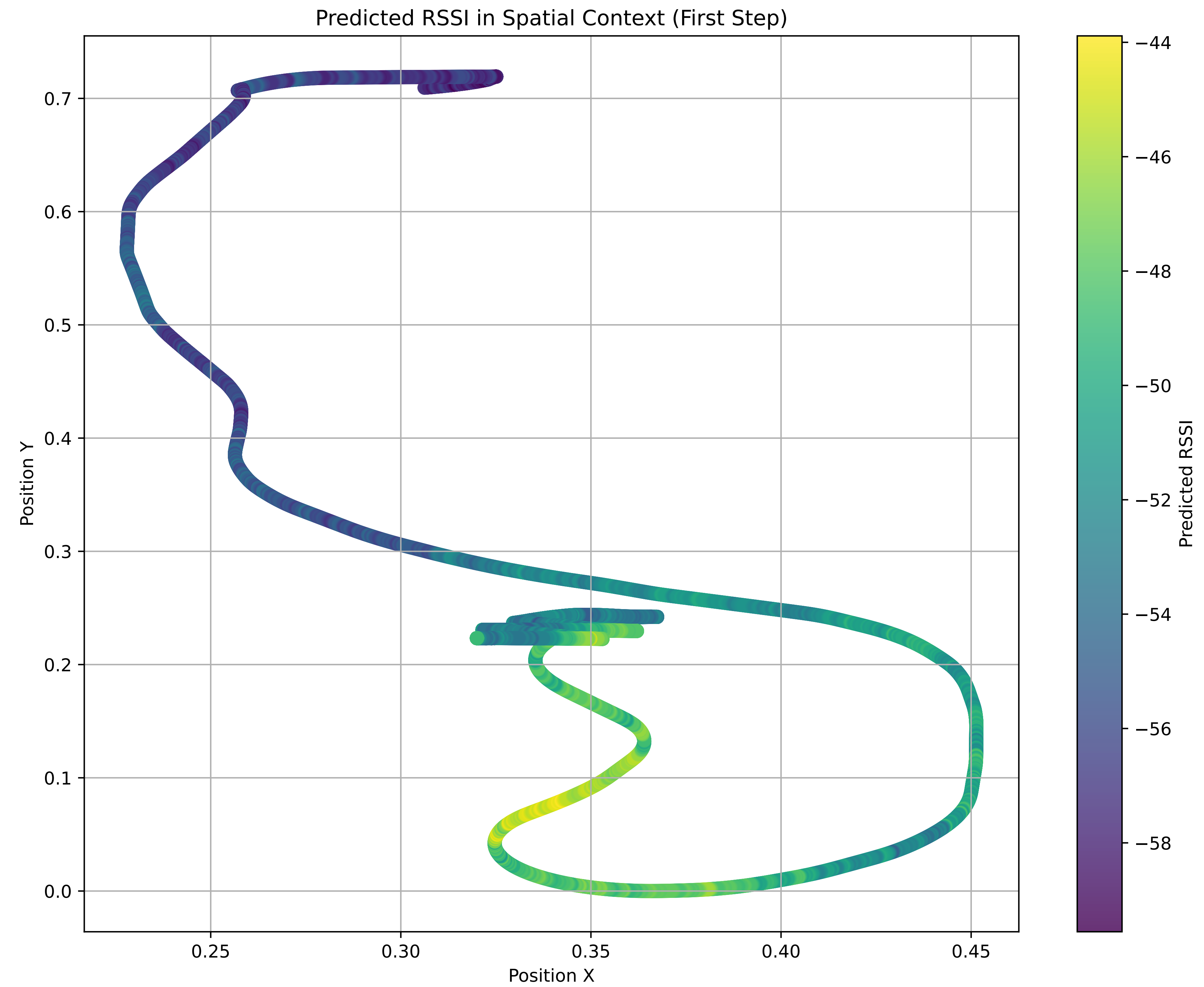

Actual vs Predicted RSSI (First Step)
Predicted RSSI
Actual RSSI
−40
−45
−50
−55
−60
−60
−55
−50
−45
−40


Actual vs Predicted RSSI (Last Step – step 20)
Predicted RSSI
Actual RSSI
−40
−45
−50
−55
−60
−60
−55
−50
−45
−40

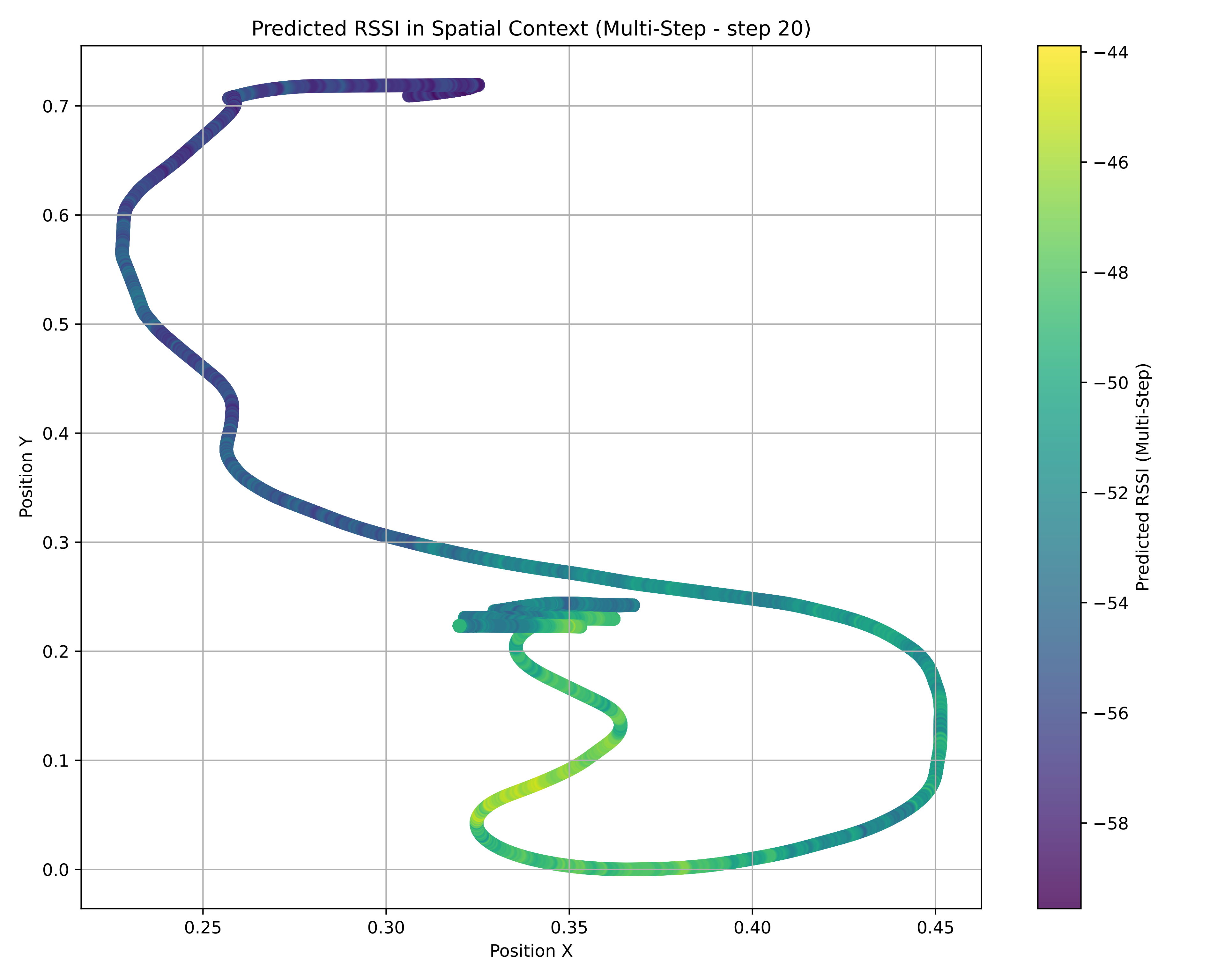

Distribution of Prediction Errors (First Step)
Frequency
1600
1400
1200
1000
800
600
400
200
0
−4
−2
0
2
4
6
Error (Actual RSSI - Predicted RSSI)

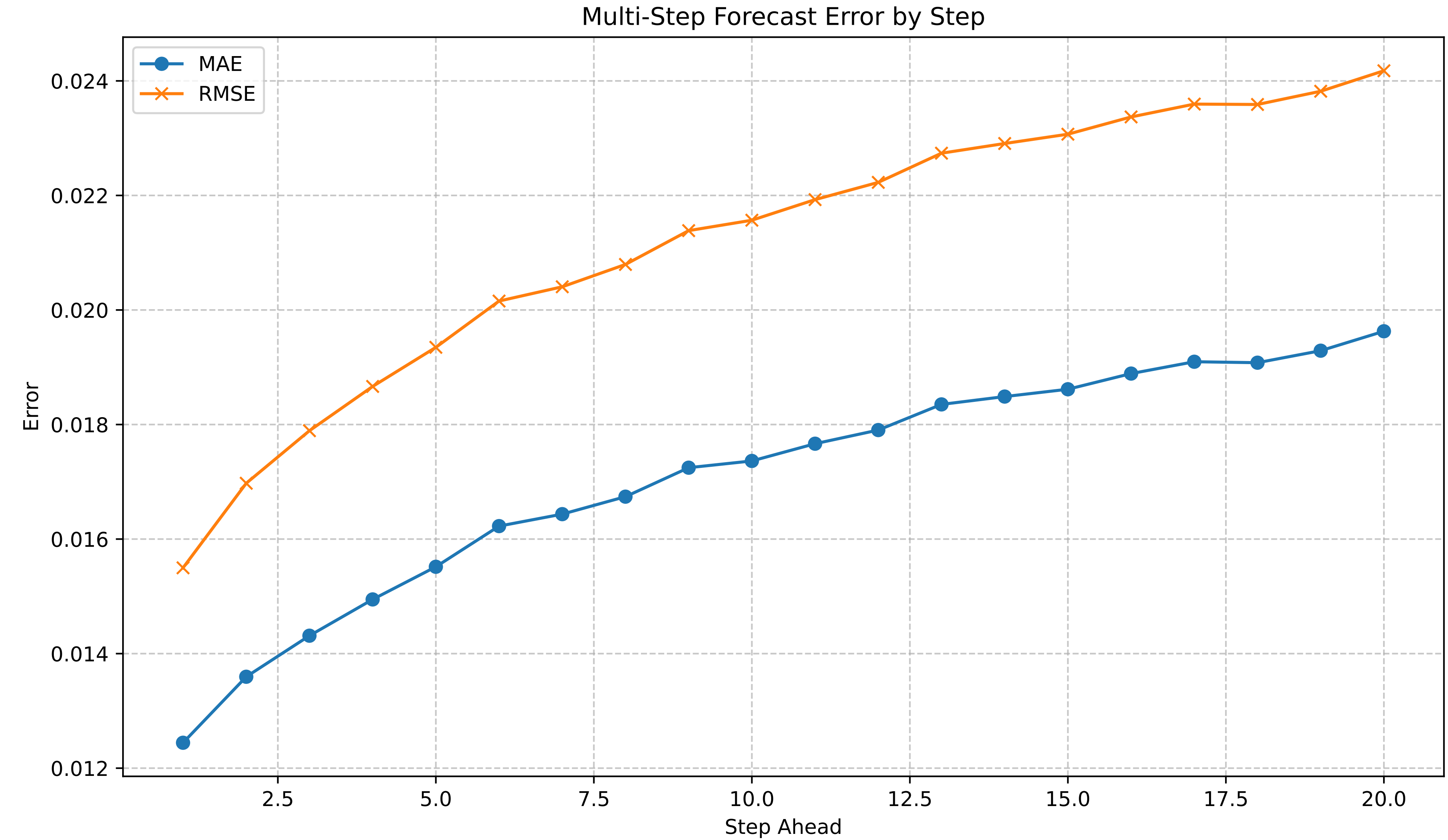


Distribution of Prediction Errors (Last Step – step 20)
Frequency
0
200
400
600
800
1000
1200
1400
−4
−2
0
2
4
6
Error (Actual RSSI - Predicted RSSI)



Predicted RSSI in Spatial Context (First Step)
Position Y
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Position X
0.25
0.30
0.35
0.40
0.45
Predicted RSSI
−44
−46
−48
−50
−52
−54
−56
−58


Predicted RSSI in Spatial Context (Multi-Step - step 20)
Position Y
Position X
Predicted RSSI (Multi-Step)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.25
0.30
0.35
0.40
0.45
−44
−46
−48
−50
−52
−54
−56
−58



Multi-Step Forecast Error by Step
MAE
RMSE
0.024
0.022
0.020
0.018
0.016
0.014
0.012
Error
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Step Ahead

```
        Resumo Automático
        -----------------
        Dataset:        rssi_odom_dataset_12apr25_ajust.parquet
        Modelo:         LSTM
        Forecast steps: 20

        MAE / RMSE (dB)
            • 1º passo : 0.01 / 0.02
            • Mediana  : 0.02 / 0.02
            • 20º passo : 0.02 / 0.02

        Crescimento do erro: 57.7%   →   Degrada Rápido (> 50 %)

        3 piores passos (MAE): 20 (0.02), 19 (0.02), 17 (0.02)

        Recomendação:
        Considere aumentar o histórico de entrada ou usar um modelo mais
profundo para melhorar multi-step.
```